

QUBO Formulation: Graph K-Coloring Problem

Graph K-Coloring Problem

Problem Statement. Given an undirected graph $G = (V, E)$ with $|V| = N$ vertices and $|E| = M$ edges, and a set of K available colors, the task is to assign exactly one color to each vertex such that no two adjacent vertices share the same color. The objective is to minimize the number of violated edges, where a violated edge is defined as an edge whose endpoints are assigned the same color. In feasibility form, a valid K -coloring exists if and only if the minimum number of violated edges is zero.

Decision Variables. We introduce binary decision variables $x_{v,c} \in \{0, 1\}$ for each vertex $v \in V$ and each color $c \in \{0, 1, \dots, K-1\}$. The variable $x_{v,c}$ equals 1 if and only if vertex v is assigned color c , and 0 otherwise.

QUBO Derivation. Step 1 – Objective Function. The number of violated edges is given by the sum over all edges of the indicator that both endpoints share a color. This yields the minimization problem:

$$\min_x H_{\text{obj}}(x) = \sum_{(u,v) \in E} \sum_{c=0}^{K-1} x_{u,c} x_{v,c}. \quad (1)$$

Since QUBO solvers minimize quadratic forms, we retain this expression directly as the objective component of the Hamiltonian.

Step 2 – Constraints. Each vertex must be assigned exactly one color. This is enforced by the equality constraints:

$$\sum_{c=0}^{K-1} x_{v,c} = 1, \quad \forall v \in V. \quad (2)$$

Step 3 – Penalty Term Expansion. We convert each equality constraint into a quadratic penalty term using a positive weight $\lambda > 0$. For a fixed vertex v , the penalty is:

$$P_v(x) = \lambda \left(\sum_{c=0}^{K-1} x_{v,c} - 1 \right)^2. \quad (3)$$

Expanding the square and applying the idempotent property of binary variables ($x_{v,c}^2 = x_{v,c}$):

$$\left(\sum_{c=0}^{K-1} x_{v,c} - 1\right)^2 = \left(\sum_{c=0}^{K-1} x_{v,c}\right)^2 - 2\sum_{c=0}^{K-1} x_{v,c} + 1 \quad (4)$$

$$= \sum_{c=0}^{K-1} x_{v,c}^2 + \sum_{\substack{c,d=0 \\ c \neq d}}^{K-1} x_{v,c}x_{v,d} - 2\sum_{c=0}^{K-1} x_{v,c} + 1 \quad (5)$$

$$= \sum_{c=0}^{K-1} x_{v,c} + \sum_{\substack{c,d=0 \\ c \neq d}}^{K-1} x_{v,c}x_{v,d} - 2\sum_{c=0}^{K-1} x_{v,c} + 1 \quad (6)$$

$$= -\sum_{c=0}^{K-1} x_{v,c} + \sum_{\substack{c,d=0 \\ c \neq d}}^{K-1} x_{v,c}x_{v,d} + 1. \quad (7)$$

Noting that the double sum over $c \neq d$ counts each unordered pair twice, we rewrite it as $2\sum_{0 \leq c < d \leq K-1} x_{v,c}x_{v,d}$. Thus, the expanded penalty contribution for vertex v is:

$$P_v(x) = \lambda \left(-\sum_{c=0}^{K-1} x_{v,c} + 2 \sum_{0 \leq c < d \leq K-1} x_{v,c}x_{v,d} + 1 \right). \quad (8)$$

Step 4 – Combined Hamiltonian. Summing the objective and all vertex penalties yields the full QUBO Hamiltonian:

$$H(x) = \sum_{(u,v) \in E} \sum_{c=0}^{K-1} x_{u,c}x_{v,c} + \lambda \sum_{v \in V} \left(-\sum_{c=0}^{K-1} x_{v,c} + 2 \sum_{0 \leq c < d \leq K-1} x_{v,c}x_{v,d} + 1 \right). \quad (9)$$

Step 5 – Q Matrix and Offset. By collecting coefficients of like terms in $H(x)$, we identify the entries of the symmetric upper-triangular QUBO matrix Q and the constant offset c . Let $n = N \cdot K$ be the total number of variables. The non-zero entries are given by:

$$Q_{(v,c),(v,c)} = -\lambda, \quad \forall v \in V, c \in \{0, \dots, K-1\}, \quad (10)$$

$$Q_{(v,c),(v,d)} = 2\lambda, \quad \forall v \in V, 0 \leq c < d \leq K-1, \quad (11)$$

$$Q_{(u,c),(v,c)} = 1, \quad \forall (u,v) \in E, c \in \{0, \dots, K-1\}, \quad (12)$$

$$c = \lambda|V|. \quad (13)$$

All other entries $Q_{i,j}$ are zero.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the exactly-one-color constraint incurs a higher energy cost than the maximum possible reduction in the objective function. Violating the constraint for a vertex v means either assigning zero colors or assigning two or more colors. In the worst

case, assigning zero colors to v removes all edge-conflict penalties incident to v , potentially reducing the objective by at most $\sum_{u:(u,v) \in E} w_{uv}$, where w_{uv} are the edge weights (here $w_{uv} = 1$). Assigning multiple colors to v increases the penalty quadratically. Therefore, choosing $\lambda > \max_{v \in V} \deg(v)$ guarantees that the penalty dominates any feasible objective improvement. In practice, a safety margin is often added; for instance, $\lambda = 10$ is chosen when the maximum degree is small, ensuring strict dominance. The total number of variables in the resulting QUBO is $n = N \cdot K$.

Correctness Argument. Minimizing $H(x)$ over $\{0, 1\}^n$ recovers the optimal K -coloring because the Hamiltonian is constructed such that $H(x) = H_{\text{obj}}(x)$ for all feasible assignments (where constraints are satisfied and penalties vanish), while $H(x) > H_{\text{obj}}(x)$ for all infeasible assignments. Specifically, any infeasible solution violates at least one vertex constraint, incurring a penalty of at least λ . Since λ exceeds the maximum possible decrease in the objective function achievable by violating a single constraint, the energy of any infeasible solution strictly exceeds the energy of the best feasible solution. Consequently, the global minimum of $H(x)$ must correspond to a feasible assignment that minimizes the original objective, i.e., a valid K -coloring with the minimum number of violated edges.

Illustrative Example. Consider a triangle graph with $N = 3$ vertices ($V = \{0, 1, 2\}$), edges $E = \{(0, 1), (1, 2), (2, 0)\}$, and $K = 3$ colors. The total number of variables is $n = 3 \times 3 = 9$. Using the variable mapping $x_{v,c} \mapsto vK + c$, the indices are 0, 1, 2 for vertex 0, 3, 4, 5 for vertex 1, and 6, 7, 8 for vertex 2. Setting $\lambda = 10$, the general formulas yield the following concrete QUBO terms:

$$H(x) = x_{0,0}x_{1,0} + x_{1,0}x_{2,0} + x_{2,0}x_{0,0} + x_{0,1}x_{1,1} + x_{1,1}x_{2,1} + x_{2,1}x_{0,1} + x_{0,2}x_{1,2} + x_{1,2}x_{2,2} + x_{2,2}x_{0,2} \\ + 10 \sum_{v=0}^2 \left(- \sum_{c=0}^2 x_{v,c} + 2 \sum_{0 \leq c < d \leq 2} x_{v,c}x_{v,d} + 1 \right). \quad (14)$$

Expanding the penalty terms for each vertex and collecting coefficients gives the explicit quadratic form:

$$H(x) = x_0x_3 + x_1x_4 + x_2x_5 + x_3x_6 + x_4x_7 + x_5x_8 + x_6x_0 + x_7x_1 + x_8x_2 \\ - 10(x_0 + x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8) \\ + 20(x_0x_1 + x_0x_2 + x_1x_2 + x_3x_4 + x_3x_5 + x_4x_5 + x_6x_7 + x_6x_8 + x_7x_8) + 30. \quad (15)$$

The corresponding Q matrix entries match the derived general formulas with $\lambda = 10$. A valid 3-coloring, e.g., $x_{0,0} = 1, x_{1,1} = 1, x_{2,2} = 1$ (all other variables 0), yields $H(x) = 0$, confirming the existence of a feasible solution with zero violated edges. Any assignment violating the exactly-one-color constraint yields $H(x) \geq 10$, verifying the penalty dominance.